Coupled effects of soil drying and salinity on soil–plant hydraulics

Mohanned Abdalla, Mutez Ali Ahmed, Gaochao Cai, Mohsen Zarebanadkauki, Andrea Carminati

https://doi.org/10.1093/plphys/kiac229 PLANT PHYSIOLOGY 2022: 190: 1228–1241

Plant Physiology®

Research Article

Coupled effects of soil drying and salinity on soil–plant hydraulics

Mohanned Abdalla,[1,2,*] **Mutez Ali Ahmed**,[1,3] **Gaochao Cai**,[1] **Mohsen Zarebanadkauki**[1] and **Andrea Carminati**[4,†]

1 Chair of Soil Physics, Bayreuth Center of Ecology and Environmental Research (BayCEER), University of Bayreuth, Bayreuth, Germany
2 Department of Horticulture, Faculty of Agriculture, University of Khartoum, Khartoum North, Sudan
3 Department of Land, Air and Water Resources, University of California Davis, Davis, California 95616, USA
4 Department of Environmental Systems Science, Physics of Soils and Terrestrial Ecosystems, Institute of Terrestrial Ecosystems, ETH Zürich, Zurich, Switzerland

*Author for correspondence: Mohanned.Abdalla@uni-bayreuth.de
†Senior author

All authors conceptualized the study. M.A. performed the experiment and analyzed the data. A.C., M.Z., and M.A. made the simulations. M.A. wrote the manuscript with contributions of M.A.A., G.C., M.Z., and A.C.

The author responsible for distribution of materials integral to the findings presented in this article in accordance with the policy described in the Instructions for Authors (https://academic.oup.com/plphys/pages/general-instructions) is Mohanned Abdalla (Mohanned.Abdalla@uni-bayreuth.de).

Abstract

Salinity and soil drying are expected to induce salt accumulation at the root–soil interface of transpiring plants. However, the consequences of this on the relationship between transpiration rate (E) and leaf xylem water potential ($\psi_{\text{leaf-x}}$) are yet to be quantified. Here, we used a noninvasive root pressure chamber to measure the $E(\psi_{\text{leaf-x}})$ relationship of tomato (*Solanum lycopersicum* L.) treated with (saline) or without 100-mM NaCl (nonsaline conditions). The results were reproduced and interpreted with a soil–plant hydraulic model. Under nonsaline conditions, the $E(\psi_{\text{leaf-x}})$ relationship became progressively more nonlinear as the soil dried ($\theta \leqslant 0.13\,\text{cm}^3\ \text{cm}^{-3}$, $\psi_{\text{soil}} = -0.08$ MPa or less). Under saline conditions, plants exhibited an earlier nonlinearity in the $E(\psi_{\text{leaf-x}})$ relationship ($\theta \leqslant 0.15\,\text{cm}^3\ \text{cm}^{-3}$, $\psi_{\text{soil}} = -0.05$ MPa or less). During soil drying, salinity induced a more negative $\psi_{\text{leaf-x}}$ at predawn, reduced transpiration rate, and caused a reduction in root hydraulic conductance (from 1.48×10^{-6} to 1.30×10^{-6} $\text{cm}^3\ \text{s}^{-1}\ \text{hPa}^{-1}$). The model suggested that the marked nonlinearity was caused by salt accumulation at the root surface and the consequential osmotic gradients. In dry soil, most water potential dissipation occurred in the bulk soil and rhizosphere rather than inside the plant. Under saline-dry conditions, the loss in osmotic potential at the root surface was the preeminent component of the total dissipation. The physical model of water flow and solute transport supports the hypothesis that a buildup of osmotic potential at the root–soil interface causes a large drop in $\psi_{\text{leaf-x}}$ and limits transpiration rate under drought and salinity.

Introduction

Plants are increasingly subjected to episodes of edaphic and atmospheric water deficit during their life spans (Choat et al., 2018; Brodribb et al., 2020). Simultaneously, plants suffer from salinity in many areas worldwide (Kumar and Sharma, 2020; Harper et al., 2021; Hopmans et al., 2021). Hence, understanding how plants react to co-occurring stressors can play a substantial role not only in stabilizing crop performance under combined stress, but also in the conservation of natural vegetation (Chaves et al., 2009).

Received March 08, 2022. Accepted April 27, 2022. Advance access publication May 17, 2022

Using soil–plant hydraulic frameworks, recent studies have endeavored to understand plant responses to water deficit in both soil and atmosphere (Sperry and Love, 2015; Carminati and Javaux, 2020; Liu et al., 2020). Soil water deficit generates hydraulic limitations to water uptake and transpiration (Draye et al., 2010; Rodriguez-Dominguez and Brodribb, 2020; Abdalla et al., 2021; Bourbia et al., 2021). In dry soils, below-ground hydraulic limitations shape the relationship between transpiration rate and leaf water potential and subsequently the onset of stress (Carminati and Javaux, 2020; Abdalla et al., 2021; Bourbia et al., 2021; Cai et al., 2021; Javaux and Carminati, 2021; Abdalla et al., 2022). Revealing the mechanism involved in shaping the relationship between transpiration rate and leaf water potential, as well as triggering stomatal closure under drought, is a prerequisite to understanding transpiration reduction under other coexisting biotic and/or abiotic stresses.

Soil salinity can limit crop growth and productivity, with great impacts in arid and semi-arid regions (Munns, 2002; Fricke et al., 2004; Munns and Tester, 2008; Chaves et al., 2009). High salt concentration in soil limits the ability of roots to take up water (Munns and Tester, 2008), similar to the impact of soil drying (Munns, 2002). Salinity triggers stomatal closure, subsequently impacting photosynthesis and plant growth and development (Munns, 2002; Fricke et al., 2004; Munns and Tester, 2008; Chaves et al., 2009). Besides reducing water availability, salt accumulation in soil induces toxic effects of sodium and chlorine ions in crops and trees (Munns and Tester, 2008; van Zelm et al., 2020; Zhang et al., 2021).

Recently, Munns et al. (2020b) postulated that plants must exclude 98% of Na^+ and Cl^- from saline soil while taking up water to avoid ion accumulation inside plant tissues (Munns and Gilliham, 2015; Munns et al., 2020b). Consequently, salts might build up around roots and possibly induce osmotic stress (Stirzaker and Passioura, 1996). Nonetheless, determining salt concentrations at the root–soil interface of intact plants during soil drying is difficult, and our understanding of salt accumulation is incomplete.

Various experimental approaches have been utilized to explore salt accumulation around the root. Hamza and Aylmore (1992a, 1992b) integrated invasive and noninvasive methods to determine water extraction, Na^+ concentration near roots, transpiration rate, and the corresponding leaf water potential (Hamza and Aylmore, 1992b). The authors showed salt accumulated at the root–soil interface in wet soils (Hamza and Aylmore, 1992a, 1992b). However, their method was prone to experimental artifacts in dry soil conditions (Hamza et al., 2001). Magnetic resonance imaging has also been utilized for noninvasive imaging of solute transport to roots (Haber-Pohlmeier et al., 2017). However, its application to quantify NaCl concentrations at the root–soil interface has been limited to specific soil types and relatively small sample sizes (Perelman et al., 2020b). Furthermore, the leaf water potential was not measured in the latter study. Stirzaker and Passioura (1996) investigated the resistances in water pathways from soil to leaf and the consequences of the osmotic stress. The authors used nonsaline nutrient solution, which mimicked the range of osmotic potential in field conditions, that is, −0.07 MPa or more (Stirzaker and Passioura, 1996). The authors concluded that osmotic stress had occurred due to salt build-up at the root–soil interface (Stirzaker and Passioura, 1996). However, the combined effects of salinity and water stress were not investigated.

Salts are transported toward the root surface due to convective and diffusive processes during root water uptake. Convection processes are driven by transpiration and the water fluxes toward the root surface and tend to increase salt content at the root–soil interface. Diffusive transport of solutes tends to equalize the concentration of salts (within the liquid phase). Solute accumulation at the root surface depends on the relative importance of convection and diffusion. As soil moisture affects both the water fluxes and the diffusion coefficient (Moldrup et al., 2001), it is a prerequisite to knowing the gradients in soil moisture around the roots, which depend on both soil hydraulic properties and transpiration rates. Therefore, to fully predict salt accumulation at the root–soil interface, one needs to couple models of water flow to the root with a model of solute transport in soils (van Lier et al., 2009; Schröder et al., 2014; Jorda et al., 2018). However, experimental validations and measured parameters for these simulations remain lacking.

We hypothesize that salt accumulation at the root surface will hinder transpiration by increasing the osmotic gradients at the root surface. Hence, a further decline in leaf water potential will be required to sustain transpiration under salinity conditions. Additionally, salt might reduce root hydraulic conductance possibly as a consequence of salt accumulation inside the root or due to sodium toxicity affecting internal root tissues (Azaizeh and Steudle, 1991; Azaizeh et al., 1992; Fricke et al., 2004; Boursiac et al., 2005). Furthermore, NaCl has been documented to impact the swelling and dispersion of clay particles within soil aggregates (Gupta and Verma, 1985), which potentially changes soil structure and pore geometry and hence impacts soil hydraulic properties (Tuller et al., 1999; Klopp and Daigh, 2020).

In this study, we combined a root pressure chamber (Cai et al., 2020b) and a model of soil water flow and salt transport to investigate the coupled effects of soil drying and salinity on soil–plant hydraulics. We measured the relationship between transpiration rate and leaf xylem water potential of tomato (*Solanum lycopersicum* L.) with and without NaCl treatment. This method accurately measures the drop in water potential at the root surface in intact plants with high temporal resolution (Carminati et al., 2017; Cai et al., 2020b; Cai et al., 2021), which can be potentially induced by salt accumulation. In parallel, we also evaluated the effect of salinity (100-mM NaCl) on the hydraulic properties of the soil. We used a soil–plant hydraulic model of Carminati and Javaux (2020) to reproduce and interpret the measurements. The model accounts for the series of resistances across the soil–plant continuum to predict transpiration rate and leaf water potential at a given soil water potential. The model was implemented to include the effect of salt accumulation at the root surface.

Results

Plants were grown in similar conditions and NaCl treatment was applied just before the drying cycle. We found no significant differences in total root length (31.9 m ± 7.9 and 37.5 m ± 3.1 standard deviation; $P = 0.323$) and root radius (0.0215 cm ± 0.0002 and 0.0219 cm ± 0.0008; $P = 0.44$) between the nonsaline and saline conditions, respectively (Supplemental Figure S1; $n = 3$).

The NaCl treatment influenced the relationship between transpiration rate and leaf xylem water potential during soil drying (Figure 1). In wet conditions, leaf xylem water potential in nonsaline plants decreased linearly with increasing transpiration rate, whereas NaCl-treated ones showed nonlinearity under relatively wet soil conditions ($\theta \leqslant 0.15\,cm^3\ cm^{-3}$; Figure 1; Table 1). In dry soils, nonlinearity emerged in both treatments, and it was more marked under saline conditions (Figure 1). Nonlinear relationship (significant quadratic term of regression, $P < 0.05$) occurred at $\theta \leqslant 0.15\ cm^3\ cm^{-3}$ under saline conditions, whereas, in nonsaline conditions, the nonlinearity (significant quadratic term of regression, $P < 0.05$) was observed at $\theta \leqslant 0.13\,cm^3\ cm^{-3}$. We have additionally fitted two linear lines to the measurements as shown in Supplemental Figures S2 and S3.

Salinity significantly reduced the maximum transpiration rate during soil drying, with and without plant pressurization (Figure 2; $P < 0.01$; Supplemental Tables S1 and S2; $n = 3$). Pressurization sustained higher transpiration rates in dry soils as well as in wet soils under salinity conditions (Figure 2). The increment in photosynthetic photon flux density (PPFD) significantly increased E during soil drying (Figure 2; Supplemental Table S2; $P < 0.001$; $n = 6$).

Predawn leaf xylem water potential deviated from soil matric potential as soil started to dry in both treatments. Nonsaline plants showed higher (less negative) predawn leaf water potential in comparison to the soil matric potential in dry soil (Figure 3). In contrast, plants with 100-mM NaCl treatment showed lower (more negative) predawn leaf water potential in wet conditions with a progressive decline as the soil dried (e.g. from −0.05 to −0.3 MPa in wet soil and from −0.2 to −0.9 MPa in dry soil; Figure 3). The analysis of covariance (ANCOVA) showed a significant shift to lower (more negative) predawn leaf water after 100-mM NaCl treatment ($P < 0.01$; $n = 3$; Supplemental Figure S4; Supplemental Tables S3 and S4). In wet soils, osmotic potential in soil was −0.02 MPa and declined to −0.29 MPa with salinity treatment. NaCl treatment showed no significant influence (confidence intervals at 95%) on either water retention curve or unsaturated hydraulic conductivity curve of the sandy loam used in this study, especially within the range of measurements (Figure 4).

The soil–plant hydraulic model reproduced the relationship between transpiration rate and leaf xylem water potential during soil drying with and without NaCl treatments (Figure 5, A and B). The modeling results showed that salt treatment induced a rapid decline in leaf xylem water potential compared to nonsaline treatment (Figure 6, A and B). The decline in leaf xylem water potential during soil drying was accelerated by salinity (Figure 6, A and B). In the model, we inversely varied root length (as a fitting parameter) to reproduce the measured relationship between leaf xylem water potential and transpiration rate. Root length active in water uptake was estimated to be 1,536 and 5,244 cm for nonsaline and saline treatment, respectively. The estimated root length must be interpreted with caution (see "Discussion"). With NaCl treatment, root hydraulic conductance ($1.30 \times 10^{-6}\ cm^3\ s^{-1}\ hPa^{-1}$) was slightly lower than the nonsaline treatment in wet soil conditions ($1.48 \times 10^{-6}\ cm^3\ s^{-1}\ hPa^{-1}$; $P = 0.692$, $n = 3$).

The modeling results suggested that, in dry and saline conditions, the salt concentration at the root surface increased by a factor of three within one day of measurements (Figure 6, C and D), which was less evident in nonsaline conditions (Figure 6C). These results revealed that salts have accumulated at the root surface under dry and saline soil conditions. Note that also in nonsaline soil a minimal build-up of solutes occurred under high transpiration rate and dry soil conditions. The decline in leaf xylem water potential temporally followed the accumulation of NaCl at the root surface, particularly with salt treatment (Figure 6). We simulated the depletion in soil water content (SWC) within the vicinity around the root (0.05–0.2 cm) during the measurements (Figure 7). In wet conditions, during measurements time (five and three hours for nonsaline and saline-treated measurements, respectively), SWC decreased from 0.23 to 0.22 $cm^3\ cm^{-3}$ and from 0.22 to 0.19 $cm^3\ cm^{-3}$ with and without NaCl treatment, respectively (Figure 7, A and B). The maximum transpiration rate under wet nonsaline conditions was $1.5 \times 10^{-3}\,cm^3\ s^{-1}$ (Figure 2A). The limited decrease of SWC in wet saline conditions was a reflection of the limited transpiration rate under these conditions (see transpiration without pressurization in Figure 2B). In dry conditions, the depletion rate was lower ($\sim 0.01\,cm^3\ cm^{-3}$; Figure 7, C–F). The corresponding maximum transpiration rate obtained without plant pressurization ($\sim 0.5 \times 10^{-3}\,cm^3\ s^{-1}$) was only one-third of that in wet conditions (Figure 2A). Note that small changes in SWC generate large gradients in soil water potential in dry conditions compared to wet conditions (see Figure 4).

Osmotic potentials (ψ_π) around roots were lower after NaCl treatment (Figure 8). The decline in ψ_π was more pronounced near the root surface after several hours of transpiration (Figure 8, A–D). In wet soils, the decrease of ψ_π was negligible in nonsaline plants; however, it declined to −0.24 MPa in saline conditions (Figure 8, A and B). In dry soils, ψ_π at the root surface was −0.2 MPa in nonsaline plants and dropped to −0.8 MPa after salt treatment (Figure 8, E and F).

The decreases in water potentials across the soil–plant continuum were estimated from modeling the experimental

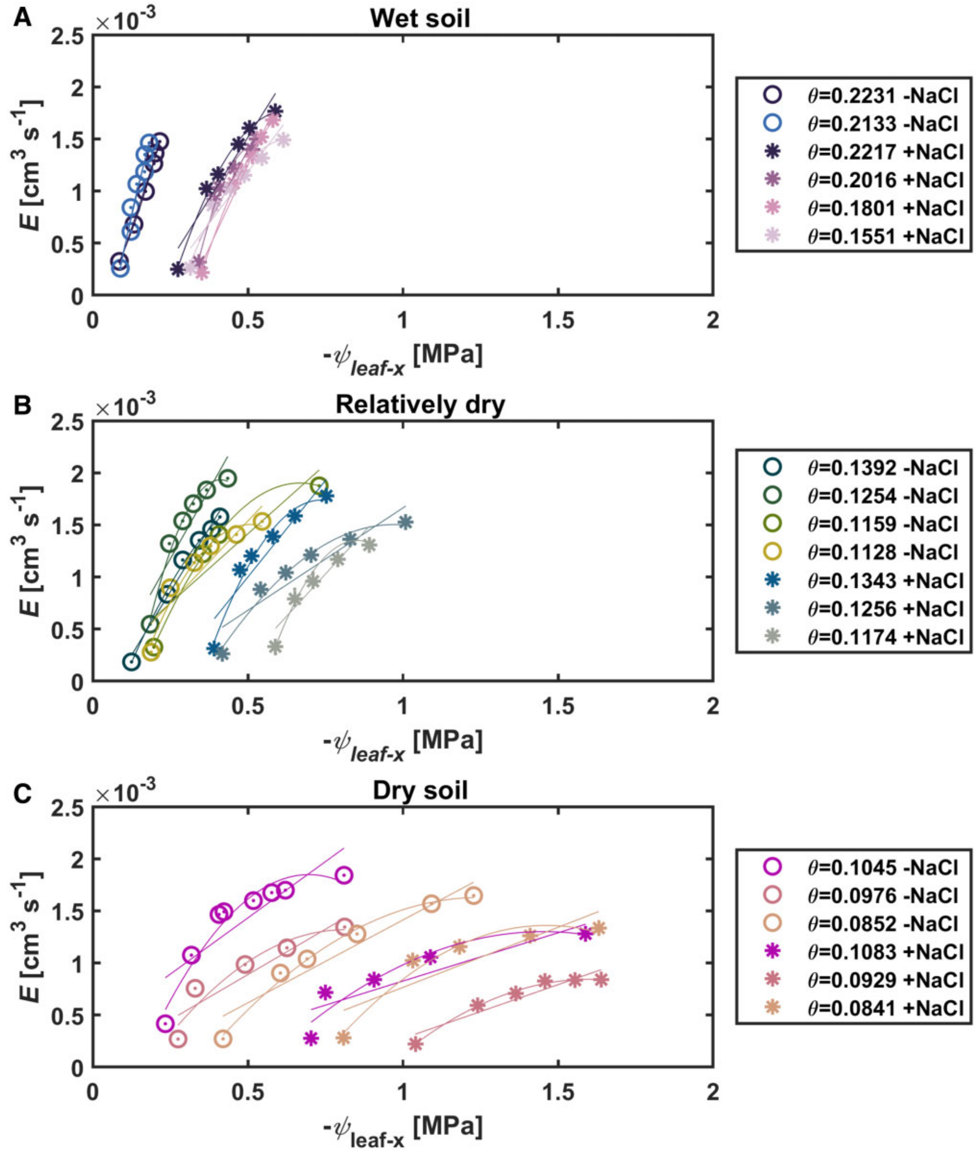

Figure 1 Relationship between transpiration (E) and leaf xylem potential ($\psi_{leaf\text{-}x}$) in nonsaline (−NaCl; open symbols) and saline-treated plants (+NaCl; asterisks) during soil drying. Without NaCl treatment, the relationship is linear in wet conditions (A), starts to bend as the soil dries (B), eventually shows clear nonlinearity in dry soils (C). Under saline conditions (i.e. 100 mM NaCl), $E(\psi_{leaf\text{-}x})$-relation shows a more marked nonlinearity, especially with high E during soil drying. The relationship was fitted, at each SWC, with linear and quadratic functions (see values of R^2 in Table 1). In sum, soil drying accentuates the impacts of NaCl. Similar colors within each subplot stand for similar SWCs (θ; $cm^3\ cm^{-3}$).

Table 1 Estimations of R^2 for linear and nonlinear (quadratic) models fitting the relationship between transpiration rate and leaf water potential under saline and nonsaline conditions in Figure 1

Saline										
SWC ($cm^3\ cm^{-3}$)	0.2217	0.2016	0.1801	0.1551	0.1343	0.1256	0.1174	0.1083	0.0929	0.0841
R^2 linear	0.92	0.90	0.97	0.91	0.87	0.86	0.81	0.77	0.88	0.75
R^2 quadratic	0.99	0.98	0.99	0.98	0.97	0.98	0.95	0.89	0.99	0.96

Nonsaline									
SWC ($cm^3\ cm^{-3}$)	0.2231	0.2133	0.1392	0.1254	0.1159	0.1128	0.1054	0.0976	0.0852
R^2 linear	0.99	0.95	0.98	0.85	0.84	0.84	0.73	0.85	0.91
R^2 quadratic	0.99	0.95	0.99	0.98	0.99	0.96	0.94	0.92	0.99

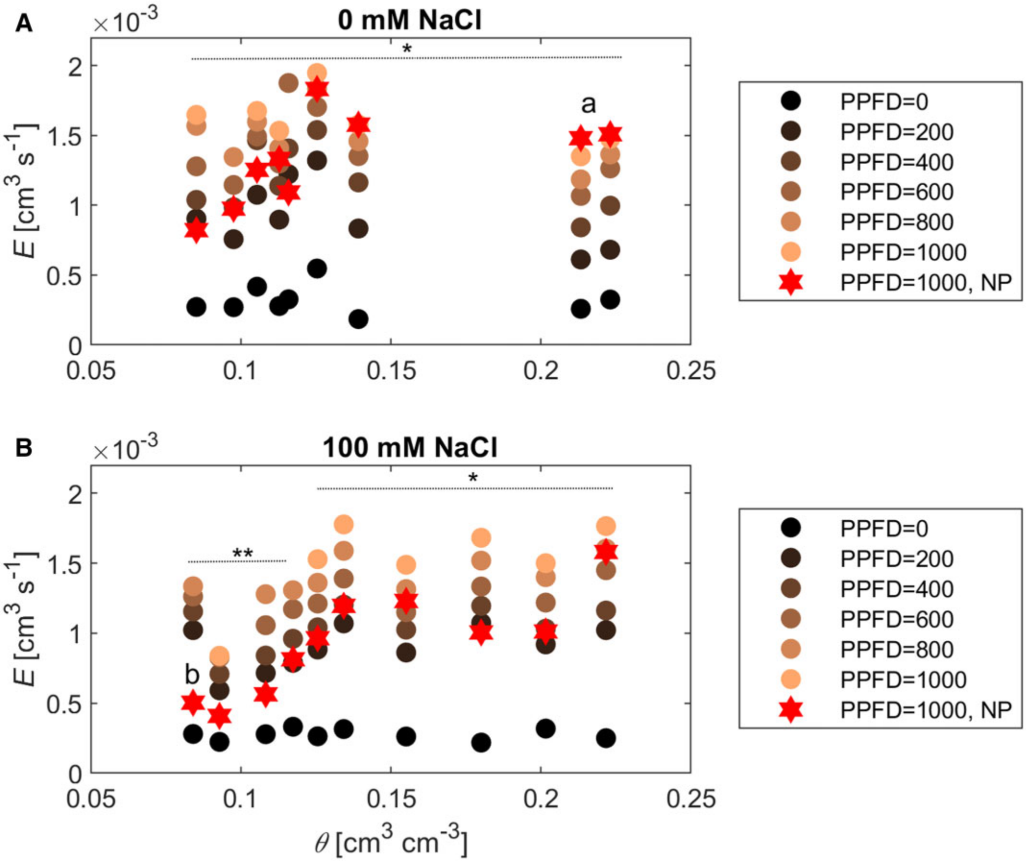

Figure 2 Transpiration (E) at different SWCs (θ) and PPFDs (μmol m^{-2} s^{-1}). A, Under nonsaline conditions and (B) under saline conditions. E was obtained with (closed symbol) and without plant pressurization (stars; NP). Salinity significantly reduced maximum E during soil drying as denoted by lowercase letters for unpressurized ($P < 0.01$, $n = 3$, Tukey–Kramer test) and asterisks for pressurized measurements ($P < 0.001$, $n = 3$, Tukey–Kramer test).

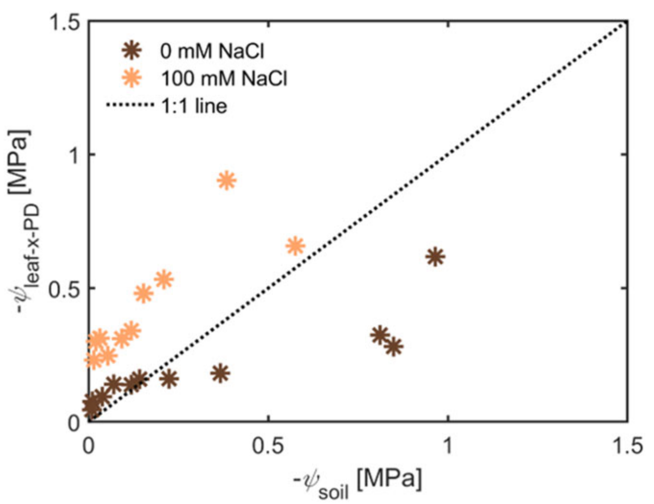

Figure 3 Impacts of salinity on predawn leaf water potential ($\psi_{leaf\text{-}x\text{-}PD}$) at different soil matric potentials (ψ_{soil}). Values of $\psi_{leaf\text{-}x\text{-}PD}$ are corresponding to the intercept of $E(\psi_{leaf\text{-}x})$-relations in Figure 1. Salinity shifted $\psi_{leaf\text{-}x\text{-}PD}$ to be more negative compared to the nonsaline treatment at similar soil matric potential ($P < 0.01$; $n = 3$; ANCOVA followed by Tukey–Kramer test; see Supplemental Figure S4 and Supplemental Tables S3 and S4).

results and are shown at the maximum transpiration rates for three exemplary SWCs. In wet conditions (SWC = 0.22), the decrease of the potential in nonsaline treatment occurred mainly in the plant due to the resistance to water flow (−0.17 MPa). Under saline conditions, the decrease in the osmotic potential was important (−0.24 MPa; Figure 9). In dry soil conditions (SWC = 0.10, 0.09), water potential dissipation in the bulk soil and the rhizosphere were greater than the loss inside the plant (Figure 9). The loss in osmotic potential at the root surface was the preeminent component of the total loss under saline-dry conditions (−0.83 MPa; Figure 9). The total loss represented leaf water potential at maximum transpiration rate. Note that these values are based on the model results. Although the model well fitted the data, the underlying assumptions did impact the results. For instance, processes such as root shrinkage and the consequent loss of conductivity in the rhizosphere are not explicitly simulated. Shrinkage is likely to have occurred in wetter soils under saline conditions due to the buildup of solutes. The model would compensate for neglecting such a process but underestimating the model parameters, such as the active root length L.

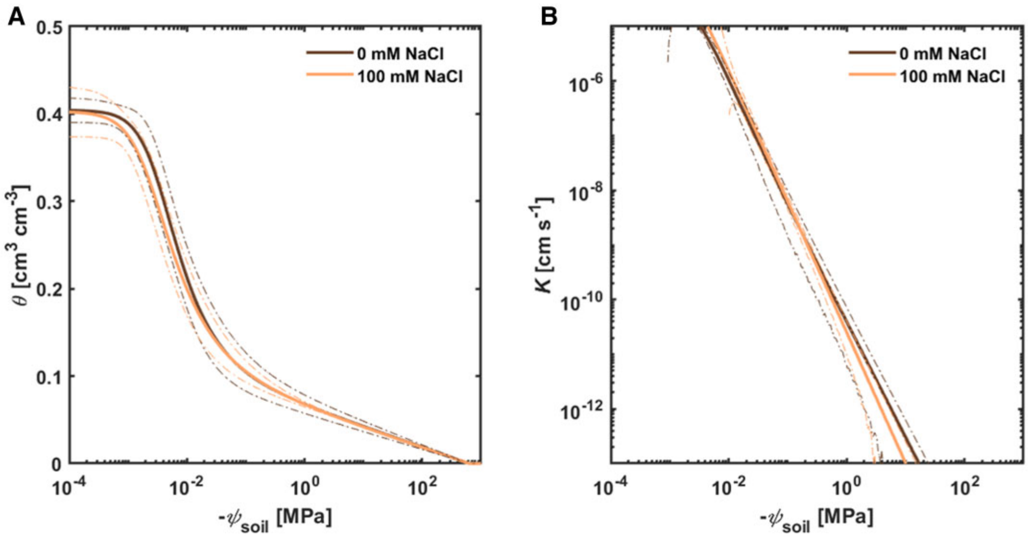

Figure 4 Soil hydraulic properties of the nonsaline (0 mM NaCl) and saline (100 mM NaCl) sandy loam soil. A, Soil water retention curves, that is, the relationship between soil water potential (ψ_{soil}) and SWC (θ). B, Soil hydraulic conductivity (K) curves. The curves were parameterized with PDI model (Peters et al., 2015). NaCl treatment did not show significant influence on soil hydraulic properties (dotted lines depict confidence intervals at 95%, n = 3), especially within the range of measurements (θ: 0.22–0.07 cm^3 cm^{-3}; K: 10^{-5} to 10^{-11} cm s^{-1}).

Discussion

We demonstrated that the combined effects of salinity and soil drying altered the relationship between transpiration and leaf xylem water potential in tomato. Without salt treatment, the relationship between transpiration rate and leaf xylem water potential was mostly linear in wet soil conditions; thereafter, as soil progressively dried, the relationship became nonlinear (Figure 1). These results are in line with previous findings in tomato (Abdalla et al., 2021), as well as other crops (Passioura, 1980; Deery et al., 2013; Carminati et al., 2017; Hayat et al., 2019; Cai et al., 2020a, 2021; Hayat et al., 2020). On the other hand, saline and dry soil entailed a more severe decline in leaf xylem water potential for increasing transpiration (Figures 1 and 5). The reduction in the slope of the $E(\psi_{leaf\text{-}x})$ relationship can be explained by the accumulation of salts at the root–soil interface and the decrease in osmotic potential at low SWCs or at high transpiration rates. Indeed, the model of solute-diffusion coupled with water flow confirmed this explanation. However, we cannot exclude the possibility that NaCl also impacts root hydraulic conductance, which might be due to salt accumulation inside the root or at the root surface (Stirzaker and Passioura, 1996). The premise is that NaCl accumulates inside the root and reduces its hydraulic conductance at the cellular level (Tyerman et al., 1989; Azaizeh and Steudle, 1991; Azaizeh et al., 1992; Wan, 2010), as well as the entire root system (Boursiac et al., 2005; Fricke et al., 2014). Under saline conditions, the decline in root conductance might potentially explain transpiration reduction even in wet soils, owing to the fact that root resistance is a considerable hydraulic limitation to transpiration (Vadez, 2014; Ahmed et al., 2018; Carminati et al., 2020; Rodriguez-Dominguez and Brodribb, 2020; Abdalla et al., 2021; Cai et al., 2022).

Predawn leaf water potential deviated from soil matric potential in both treatments. In nonsaline treatment, predawn leaf water potential was higher than that of the soil. However, the soil matric potential, as obtained from time-domain refractometer (TDR) and soil retention curve, is an approximation of what the plant senses. Furthermore, the water retention curve was measured in unplanted soil, and plants might change this relationship (Helliwell et al., 2017). Additionally, the leaf xylem might have had a greater solute concentration than the soil in the nonsaline treatment, which would explain the difference. Salinity induced a more negative predawn water potential in saline compared to nonsaline conditions (Figure 3), which is well explained by the low osmotic potential of the soil (Donovan et al., 2001; Cai et al., 2020a; Zhang et al., 2021).

Na^+ and Cl^- exclusion during water uptake is proposed as a successful mechanism of salt tolerance in crops (Munns, 1985; Munns and Tester, 2008; van Zelm et al., 2020). Subsequently, salt concentration near roots will increase compared to bulk soils (Munns et al., 2020a; Munns et al., 2020b; Perelman et al., 2020a). This accumulation induces additional osmotic gradients at the root surface in dry soils, as predicted by the model (Figures 6, C, D and 8). Generally, ions that are not excluded from the transpiration stream might accumulate inside plant tissues causing irreversible damage (Passioura, 2020), especially under prolonged exposure to saline conditions. Plant responses to salinity occur over a wide time range, starting from the first minutes of exposure up to several months in perennial species (Munns, 2002). Short-term exposure (1–3 days) was criticized as a nonsufficient time course allowing plant adaptation to salinity (Passioura, 2020). In the current study, the plants experienced salinity for 2–3 days before starting the measurements, which lasted circa 7–9 days, implying

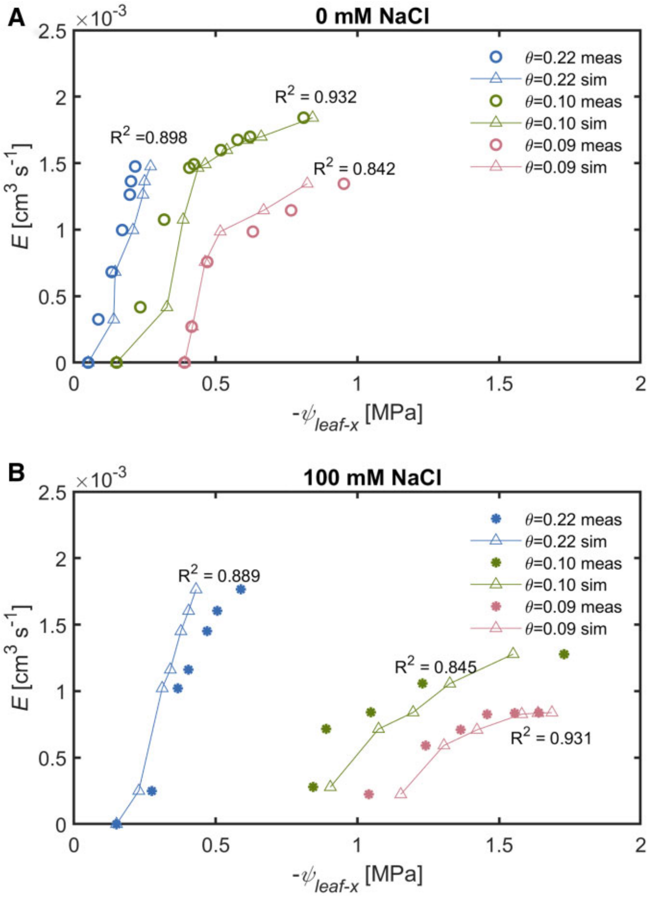

Figure 5 Comparison of transpiration rate (E) and leaf xylem water potential ($\psi_{\text{leaf-x}}$) between nonsaline and saline conditions. The relationship between E and $\psi_{\text{leaf-x}}$ was reproduced by the soil–plant hydraulic model for (A) nonsaline and (B) saline-treated plants during soil drying. Water stress level was represented at three SWCs (θ = 0.22, 0.10, and 0.09 cm^3 cm^{-3}) for both nonsaline and saline-treated plants (meas, measured; sim, simulated).

that either Na^+ and Cl^- exclusion was efficient and ions' concentrations remained below toxic levels or plants started to adapt and compartmentalized the ions within vacuoles. Prolonged exposure (>8 days) to high concentrations of NaCl was reported to induce lethal consequences on leaves of barley (*Hordeum vulgare* cv. Clipper; Munns and Passioura, 1984). In our study, out of the previously mentioned ~7 days, each plant was pressurized for 3 or 4 days and only during daytime hours. Thus, salt-related responses (i.e. accumulation and root adaptation, especially within the context of soil–plant hydraulics) found sufficient time to occur, similarly to anticipated processes in field conditions in a comparable time frame (cf. Munns et al., 2000; Munns, 2002).

Applying pressurized gas to root and soil had kept the shoot turgid, which allowed plants to sustain higher transpiration rates, especially in dry soil conditions (Figure 2). Interestingly, in wet and saline conditions, maximum transpiration was lower without plant pressurization, suggesting that an increase in salinity-initiated stress reduced maximum transpiration by decreasing root hydraulic conductance (from 1.48 × 10^{-6} to 1.30 × 10^{-6} cm^3 s^{-1} hPa^{-1}), which supports our hypothesis. These results are in line with previous findings of Boursiac et al. (2005) and Fricke et al. (2014). Another possible explanation is that sodium ion toxicity might have occurred in leaves (due to increased transpiration/root water uptake) and induced osmotic damage on leaf tissues (Li et al., 2021; Zhang et al., 2021). Limited water depletion under saline wet conditions was also captured by the model (Figure 7B), which additionally supports the finding regarding reduced transpiration.

The model estimated an increase in the root length active in water uptake (L) under saline conditions compared to nonsaline. It should be noted that L is a fitting parameter that controls the decreases in soil water potential around the root. Thus, this apparent increase in L under salt should be considered with caution (Abdalla et al., 2021; Cai et al., 2021). It is noteworthy that we used a simplified single root model, and the use of an architecture model has shown a decrease in effective root length in dry soils (Landsberg and Fowkes, 1978). In a previous study, Hayat et al. (2019) used both simplified and root architecture models to reproduce the relationship between transpiration rate and leaf water potential under heterogeneous soil moisture conditions. The authors found that the simplified model of root water uptake was capable of reproducing the relationship in wet and dry soils (Hayat et al., 2019). The drawback of using root architecture models is that they require additional parameters that are challenging to measure, especially in intact plants.

Sodium ions were found to impact the swelling and dispersion processes of clay particles within soil aggregates (Gupta and Verma, 1985; Klopp and Daigh, 2020), resulting in a different pore geometry. However, our data suggest that this was not the case for the sandy loam used in this study and hence salinity in soil solution did not impact the hydraulic properties of bulk soil (Figure 4). It is worth noting that substrates used in this study were not aggregated soil due to sieving through a 1-mm sieve. The absence of soil aggregates might explain the neutral effect of salinity on hydraulic properties. However, our data cannot exclude the possibility that NaCl can impact the hydraulic properties of soils with diverse textures and structures.

In conclusion, salinity impacted the relationship between transpiration and leaf water potential in a predictable way. According to our modeling exercise, limited transpiration rates are explained by the accumulation of salts at the root–soil interface. Complementary measurements of salt concentration at the root–soil interface would be needed to support this interpretation.

Materials and methods

Soil and plant preparation

Grafted tomato (*S. lycopersicum* L.) plants were used in this study. Grafting facilitates the coupling of independent genotypes based on desirable traits of roots and shoots (Abdelhafeez et al., 1975; Albacete et al., 2015; Rouphael

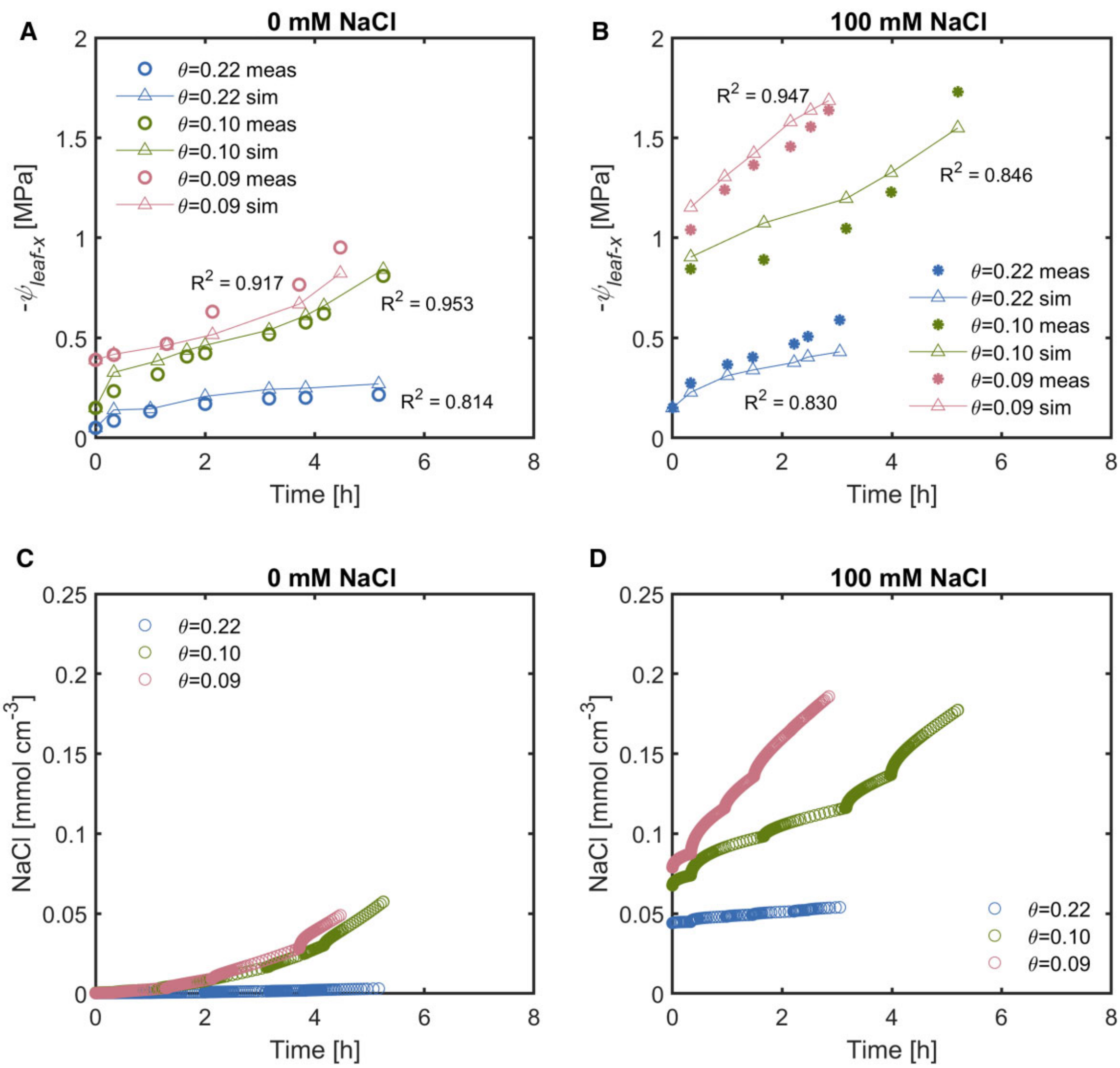

Figure 6 Changes in leaf xylem water potential ($\psi_{\text{leaf-x}}$) and NaCl over time. The dynamics of $\psi_{\text{leaf-x}}$ over time (A and B) follow salt accumulation at root surface for the corresponding time periods (C and D). Open symbols and asterisks denote measurements of nonsaline and 100-mM NaCl treatment, respectively (A and B). Connected triangles stand for the simulated values (A–B; meas: measured; sim: simulated). Salt accumulation was simulated by the soil–plant model in nonsaline (C), and saline conditions (D).

et al., 2018). Scions from the M82 variety were grafted onto wild tomato (*Lycopersicon hirsutum*) that provided rootstock, which was suggested to potentially tolerate salinity stress (Bolarín et al., 1991). Seeds of scion and rootstock were sown simultaneously in different pots. Rootstock seeds were grown in polyvinyl chloride cylinders 30 cm in height and 10 cm in diameter. Five holes with a diameter of 5 mm were made on the cylinders' side every 5 cm for SWC measurement. The cylinders were filled with sandy-loam soil with a bulk density of 1.4 g cm^{-3}. The soil was a mixture of 62.5% loamy soil and 37.5% quartz sand passing through a 1-mm sieve to facilitate homogeneity among replicates. After soil filling, the cylinders were capped with an aluminum plate with a central hole of 1.4 cm in diameter using silicon rubber glue (Teroson, Henkeln, Germany).

Seeds were grown at a depth of 1.5 cm in the central hole of the aluminum plate. Seedlings of scions and rootstocks were grafted when the diameters of their stems matched, which occurred approximately 1 week after germination. Immediately after grafting, plants were placed inside a chamber at ca. 95% relative humidity and 200-μmol m^{-2} s^{-1} PPFD for ca. 5 days. Thereafter, the established plants were placed in a climate-controlled room with a photoperiod of 14 h, a day/night temperature of 28°C/18°C, and a day/night relative humidity of 57%/65%. The PPFD was ca. 600 μmol m^{-2} s^{-1} (Luxmeter PCE-174, Meschede, Germany). SWC was maintained above 0.15 cm^3 cm^{-3} during the growth.

Two weeks after grafting, plants were transferred to the laboratory (PPFD: 200 μmol m^{-2} s^{-1}; temperature 25°C/18°C) and sealed at the collar with a glue (Supplemental Figure S5; UHU plus Endfest 300, Bühl, Germany). To introduce saline conditions, three plants (out of six) were treated with 100-mM NaCl solution. Each plant received 200 mL within 2–3 days before the start of measurements. During plant growth, SWC (θ) was measured using TDR (E-Test, Lublin, Poland). Soil osmotic potential was measured in wet

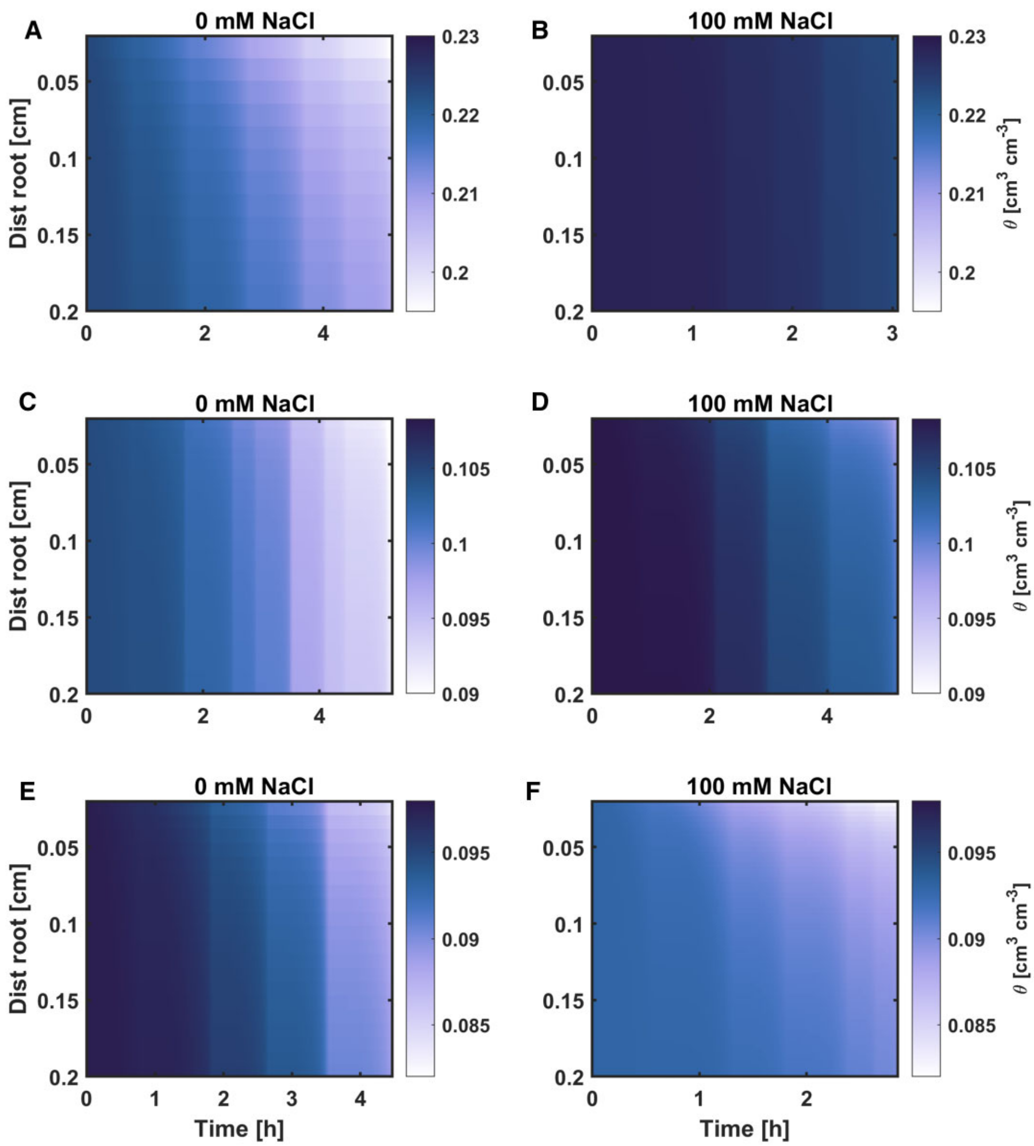

Figure 7 Spatiotemporal distribution of SWC (θ; $cm^3\ cm^{-3}$) toward the root surface. The parts show three levels of water stress, namely, $\theta \approx 0.22\ cm^3\ cm^{-3}$ for (A–B), $0.10\ cm^3\ cm^{-3}$ for (C and D), and $0.09\ cm^3\ cm^{-3}$ for (E and F). The gradients in θ are more pronounced in wet soil (when the transpiration was greater) without NaCl treatment (which had a smaller root length active in water uptake). In dry soils and salt-treated ones, the gradients are less marked, reflecting (1) an impeded water uptake (2) the shape of the water retention curve. In these figures, color bars depict θ.

unplanted soil using an osmometer (VAPRO Pressure Osmometer 5600, ELITechGroup, USA).

At the end of the experiments, plant biomass was evaluated in terms of leaf area and root length measurements. Leaf area was determined using ImageJ software (1.50e, http://imagej.nih.gov/ij). After the experiments, roots were collected and the total root length and diameter were obtained using WinRihzo software (Regent Instruments Inc., Canada).

Transpiration and leaf xylem water potential measurements

A root pressure chamber system (RPCS) was used to simultaneously measure transpiration (E) and leaf xylem water potential ($\psi_{leaf\text{-}x}$) in intact plants during soil drying (Passioura, 1980). The RPCS was recently described in Cai et al. (2020a) and Abdalla et al. (2021). Briefly, the pot was placed inside a root pressure chamber, with the shoots enclosed in a cuvette to obtain E. Light-emitting diodes were attached to the cuvette vertically to provide a PPFD for E regulation. A continuous airflow with a velocity of $8.25\ L\ min^{-1}$ passed through the cuvette. Combined temperature–humidity sensors (Galltec-Mela, Bondorf, Germany) measured the temperature and the relative humidity of the inward and the outward air continuously. E was determined by multiplying the airflow rate by the difference between the outward and the inward humidity.

The basic concept of root pressure chambers was to balance the suction inside the plant by applying pneumatic pressure to the soil and roots. The applied pressure

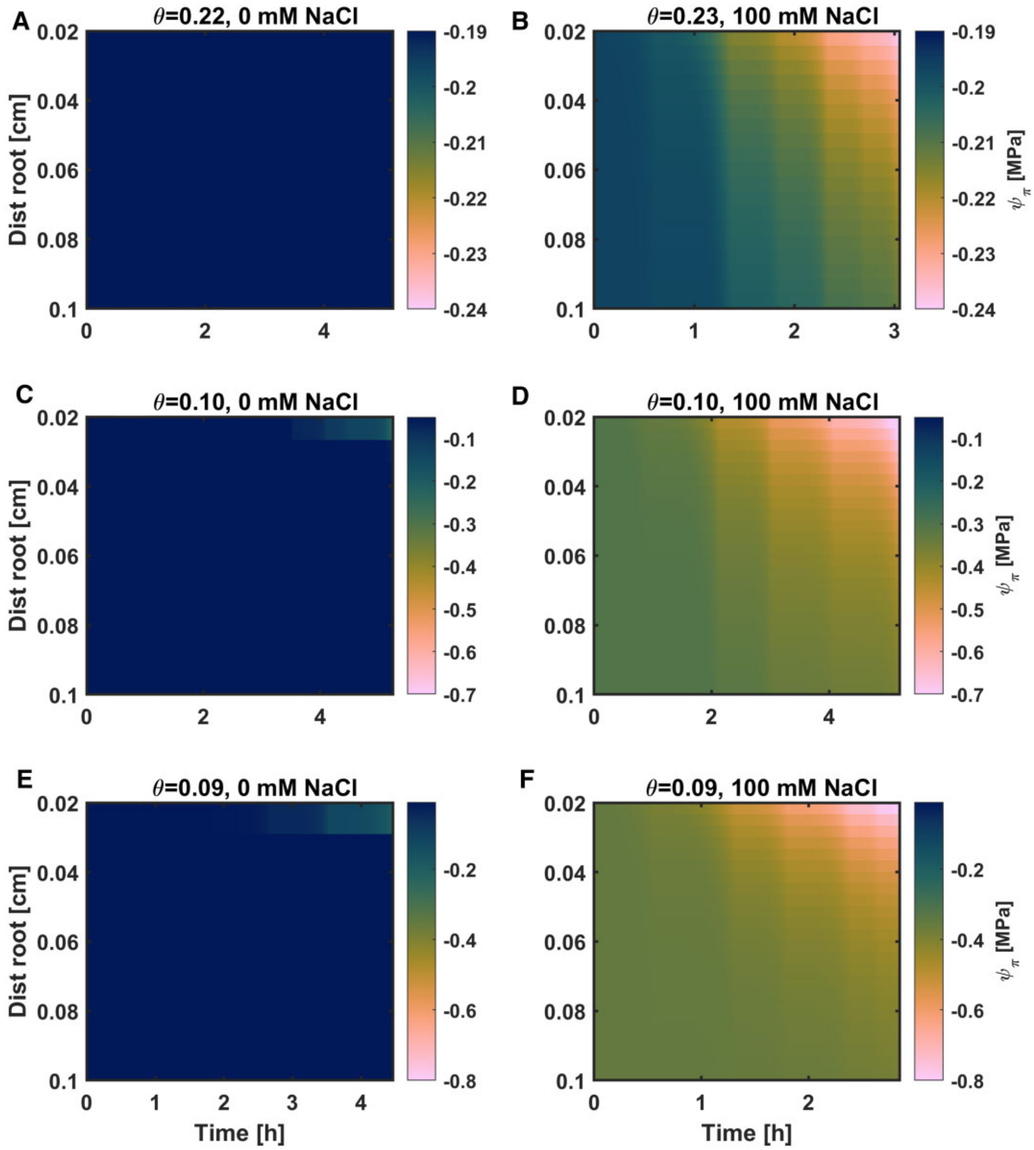

Figure 8 Spatiotemporal distribution of soil osmotic potential (ψ_π; MPa) toward the root surface. Initial SWC for (A) and (B) θ = 0.22, (C and D) θ = 0.10, (E and F) θ = 0.09 $cm^3\ cm^{-3}$. The gradients of ψ_π were originated from salt accumulation at the root surface (Figure 6, C and D), and the corresponding profile of SWC in Figure 7. Salt treatments induced additional osmotic gradients at the root surface. Here, color bars stand for ψ_π gradients and have three scales based on SWC. Please note the different time scales on the x-axis.

(balancing pressure; *P*) is numerically equal to leaf xylem tension before pressurization (Passioura, 1980; Abdalla et al., 2021). A meniscus system was connected to a leaflet cut (at the petiolule) to evaluate the stability of the droplet. The $\psi_{leaf\text{-}x}$ was determined when the meniscus was kept stable for *ca.* 10 min (Cai et al., 2020a).

E was changed by increasing the PPFD stepwise from 0 µmol $m^{-2}\ s^{-1}$ to 200, 400, 600, 800, and 1,000 µmol $m^{-2}\ s^{-1}$. Concomitantly, the corresponding $\psi_{leaf\text{-}x}$ was determined at each PPFD. Furthermore, *E* was measured at the highest PPFD (1,000 µmol $m^{-2}\ s^{-1}$) without plant pressurization. Each plant was measured at different SWCs from wet (first day) to dry (third/fourth day). Soil drying happened naturally during the days of measurements. In total, plants experienced salinity conditions for ca. 7–9 days.

Modeling water and solute flow in the soil–plant continuum

We used a model coupling water flow and solute transport to reproduce the measured relationship between transpiration rate and leaf xylem water potential. The mass conservation principle combined with the Richards equation for 1-D axisymmetric flow toward a single root, according to van Lier et al. (2006), can be written as:

$$\frac{\partial\theta}{\partial t} = \frac{C(\psi_m)\partial\psi_m}{\partial t} = -\frac{\partial(rq)}{\partial r} \tag{1}$$

$$q = -K_s\ (\psi_m)\frac{\partial\psi_m}{\partial r} \tag{2}$$

where θ is the volumetric SWC ($cm^3\ cm^{-3}$), *C* is the specific soil water capacity ($\partial\theta/\partial\psi_m$, cm^{-1}), *q* is the water

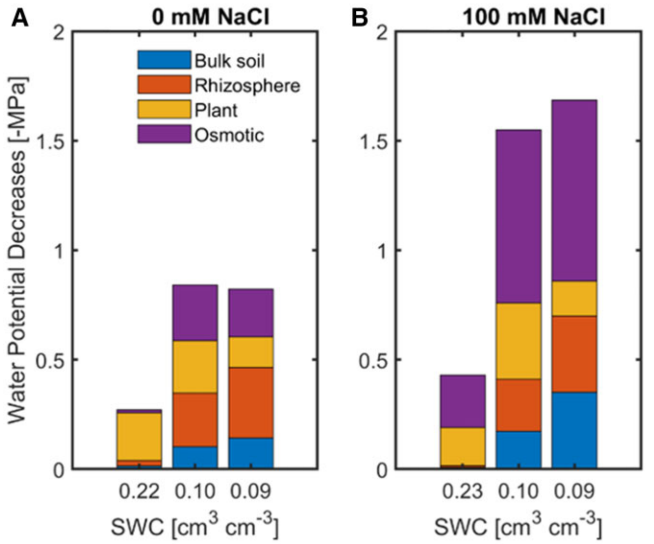

Figure 9 Dissipation of water potentials across the soil–plant continuum, divided into its components. A, Nonsaline and (B) saline conditions. Under low SWC (SWC = 0.10, 0.09), greater decreases in matric potentials occurred in the bulk soil and the rhizosphere compared to wet soil conditions (SWC = 0.22). Under saline conditions (100-mM NaCl), the decreases in soil osmotic potential were the greater contributor to the total decease, especially in dry soils.

flux (cm s^{-1}), K_s is the soil hydraulic conductivity (cm s^{-1}), ψ_m is the function of the soil matric potential (hPa, 1 hPa $\approx$ 1 cm), r is the radial distance (cm), and $\frac{\partial \psi_m}{\partial r}$ is the gradient in matric potential and t is time (s). This equation (Equation 2) was solved numerically using a fully implicit finite difference method in MATLAB software (Math Works, Inc., Natick, MA, USA). Hereafter, this equation was subjected to the following initial and boundary conditions:

$$\psi_m(t) = \psi_{m,0} \; t = 0 \tag{3}$$

$$q(r) = \frac{E}{2\pi r_0 L} \; r = r_0 \tag{4}$$

$$q(r) = 0 \; r = r_b \tag{5}$$

where $\psi_{m,0}$ is the soil matric potential at time zero and corresponding to the average SWC in the sample (cm), r_0 and r_b are the root radius and the exterior radius of soil around the root (cm), E is the transpiration rate (cm^3 s^{-1}), L is the root length active in water uptake (cm), which was used as fitting parameters. In this model, L is a single root that represents the root system. The r_b is determined by L and the volume of the pot V (cm^3), according to:

$$r_b = \sqrt{\frac{V}{\pi L}} \tag{6}$$

The solution of Equation (1) gives the profile of soil matric potential as a function of radius from the root surface. We simulated NaCl transport and its concentration in the soil solution to include the effect of osmotic potential on the leaf xylem water potential. According to van Lier et al. (2009), the combined mass conservation principle and solute transport processes under 1-D axisymmetric flow conditions give:

$$\frac{\partial(\theta + b)C_w}{\partial t} = \frac{\partial}{r\partial r}\left(\theta r D(\theta)\frac{\partial C_w}{\partial r}\right) - \frac{\partial}{r\partial r}(rqC_w) \tag{7}$$

where C_w is the concentration of NaCl in the soil solution (mol mL^{-1}), b is the coefficient describing the adsorption of Na by the solid phase (–), and $D(\theta)$ is the diffusion coefficient of Na in the soil solution (cm^2 s^{-1}) described as:

$$D(\theta) = D_0 \frac{\theta^{10/3}}{\phi^2} \tag{8}$$

where D_0 is self-diffusion coefficient of Na in free water (cm^2 s^{-1}), and ϕ is soil porosity (–). Equation (7) was solved numerically using a fully implicit finite difference method in MATLAB. Therefore, this equation was subjected to the following initial and boundary conditions:

$$C_w(t) = C_{w,0} \; t = 0 \tag{9}$$

$$\theta r D \frac{\partial C_w}{\partial r} - rqC_w = 0 \; r = r_0 \tag{10}$$

$$\theta r D \frac{\partial C_w}{\partial r} - rqC_w = 0 \; r = r_b \tag{11}$$

where $C_{w,0}$ is the concentration of Na in the soil solution (mol mL^{-1}) at time zero. Equations (10) and (11) impose a zero solute flux boundary condition at both ends of the soil domain (r_0 and r_b).

The solution of Equation (7) gives the profile of solute concentration in the soil solution as a function of both distance from root surface and time. We used the following relationship to estimate the osmotic potential (ψ_π [bar]) in the soil solution based on the profile of concentrations:

$$\psi_\pi = -\omega n R C_w T \tag{12}$$

where ω is a unit-less osmotic coefficient depending on the type of solute (ω = 0.9 for the case of NaCl), C_w is the molar concentration of Na in the soil solution (mol mL^{-1}), n is the number of ions produced when the solute undergoes dissociation ($n = 2$ for NaCl), R is the universal gas constant (0.0831 L bar mol^{-1} K^{-1}), and T is the temperature (Kelvin). The units were converted consistently in the model.

Combination of Equations (1)–(12) gives the profile of water potential in the soil as a function of distance from the root surface. In the pressure chamber experiment, we measured leaf xylem water potential at varying transpiration rates for varying SWCs. The slope of the linear part of $E(\psi_{leaf\text{-}x})$ depicts plant hydraulic conductance (K_{plant}), which is the harmonic mean of the stem xylem hydraulic conductance (K_x) and root hydraulic conductance (K_{root}). According to Abdalla et al. (2021), K_{plant} is approximately equal to K_{root}. Knowing transpiration rate and K_{root}, we estimated the dissipation of water potential between the soil–

root interface and the plant leaf xylem. Based on this principle, the water potential in the leaf xylem is given by:

$$\psi_{leaf-x} = -\frac{E}{K_{root}} + \psi_m(r_0) + \psi_\pi(r_0) \tag{13}$$

where K_{root} is the hydraulic conductance of root ($cm^3\ s^{-1}\ hPa^{-1}$) and $\psi_m(r_0)$ and $\psi_\pi(r_0)$ are the matric potential and osmotic potential in the soil at the root surface.

The difference between soil and plant osmotic potential leads to a deviation between predawn leaf and soil water potential (Donovan et al., 2001; Cai et al., 2020a; Abdalla et al., 2022), which was added to the simulations as an offset.

Model parameterization

The hydraulic properties of the soil mixture used in this study were measured via the evaporative method implemented in the Hyprop (Meter Group, Munich, Germany). This device encompasses two tensiometers and a balance, it measures gravimetric SWC and the corresponding matric potential in soil samples as described hereafter. Soil samples were packed into a Hyprop sample holder (area of 50 cm^2 and height of 5 cm) at the same bulk density as the one achieved in the cylinders used for plant growth. Three samples were saturated with water, and the other three samples were with 100-mM NaCl solution from the bottom. The samples were let dry from the top via evaporation, and the changes in SWC, matric potential at two depths (−1.25 and −3.75 cm) were recorded over time. Water retention curves and unsaturated hydraulic conductivity curves were parameterized according to the Peter–Durner–Iden (PDI) model (Peters et al., 2015). The parameters were estimated by fitting the obtained measurements and solving the Richards equation.

The K_{root} was obtained as the slope of the linear part of $E(\psi_{leaf-x})$ at each SWC. Thus, dynamic K_{root} was used during the simulation. The root length active in water uptake (L) was a fitting parameter in the model. L value was obtained from inverse modeling of measured $E(\psi_{leaf-x})$-relation in different SWCs, for both treatments. The inverse modeling approach minimized the difference between measured and simulated ψ_{leaf-x} at each E value systematically. The following objective function (ObjFunc) was used:

$$\text{ObjFunc} = \sum_{i=1}^{i=n} \frac{\left(\psi_{leaf-x}(i) - \psi_{leaf-x,\ sim}(i)\right)^2}{\psi_{leaf-x}(i)^2} \tag{14}$$

Statistical analysis

The influences of salinity and drought on transpiration, with and without plant pressurization, were statistically evaluated using N-way analysis of variance (ANOVA) followed by multiple comparison (Tukey–Kramer test). MATLAB (Math Works Inc., USA) was used to perform the analyses (N-way ANOVA: *anovan*; multiple comparison: *multcompare*). The differences in measured root length and diameter between treatments were evaluated using *t*-test. Statistical significance was considered when P-value was < 0.05. Furthermore, we tested the values of predawn leaf water potential as a function of soil water potential for both treatments using ANCOVA. We estimated the confidence intervals at 95% to identify the influences of NaCl solution on soil retention curves and soil unsaturated hydraulic conductivity curves. The relationship between transpiration rate and leaf xylem water potential at a given soil moisture was fitted with both linear and quadratic functions to test the onset of hydraulic nonlinearity statistically.

Supplemental data

The following materials are available in the online version of this article.

Supplemental Figure S1. Root parameters were similar in controlled and salt-treated plants.

Supplemental Figure S2. Relationship between transpiration rate and leaf xylem water potential in different SWCs under nonsaline conditions.

Supplemental Figure S3. Relationship between transpiration rate and leaf xylem water potential in different SWCs under saline conditions.

Supplemental Figure S4. Linear fit of the relationship between predawn leaf water potential and soil water potential.

Supplemental Figure S5. Tomato plant (*S. lycopersicum* L.) glued at the collar.

Supplemental Table S1. ANOVA considering the influences of salinity, soil drying, and their interactions on transpiration rate measured without plant pressurization.

Supplemental Table S2. ANOVA considering the influences of soil drying, salinity, PPFD and their interactions on transpiration rate measured with plant pressurization.

Supplemental Table S3. Coefficient estimates for the relationship between predawn leaf water potential and soil water potential in control and salt treatments in terms of slope and the intercept.

Supplemental Table S4. ANOVA to identify the significant difference between the slope and the intercept of the relationship between predawn leaf water potential and soil water potential in control and salt treatments as shown in Supplemental Table S3.

Acknowledgments

We thank Nimrod Schwartz for providing the seeds, Johanna Pausch for the climate chamber and WinRhizo, and Andreas Kolb for his technical assistance.

Funding

The doctoral position of Mohanned Abdalla was funded by the German Academic Exchange Service (Deutsche Akademische Austauschdienst (DAAD)). The position of Gaochao Cai was funded by the Bundesministerium für Bildung und Forschung (BMBF), Project 02WIL1489 (Deutsch-Israelische Wassertechnologie-Kooperation).

Conflict of interest statement. The authors declare no conflict of interest.

References

Abdalla M, Ahmed MA, Cai G, Wankmüller F, Schwartz N, Litig O, Javaux M, Carminati A (2022) Stomatal closure during water deficit is controlled by below-ground hydraulics. Ann Bot **129**: 161–170

Abdalla M, Carminati A, Cai G, Javaux M, Ahmed MA (2021) Stomatal closure of tomato under drought is driven by an increase in soil–root hydraulic resistance. Plant Cell Environ **44**: 425–431

Abdelhafeez AT, Harssema H, Verkerk K (1975) Effects of air temperature, soil temperature and soil moisture on growth and development of tomato itself and grafted on its own and egg-plant rootstock. Sci Hortic **3**: 65–73

Ahmed MA, Passioura J, Carminati A (2018) Hydraulic processes in roots and the rhizosphere pertinent to increasing yield of water-limited grain crops: a critical review. J Exp Bot **69**: 3255–3265

Albacete A, Martínez-Andújar C, Martínez-Pérez A, Thompson AJ, Dodd IC, Pérez-Alfocea F (2015) Unravelling rootstock × scion interactions to improve food security. J Exp Bot **66**: 2211–2226

Azaizeh H, Gunse B, Steudle E (1992) Effects of NaCl and CaCl2 on water transport across root cells of maize (Zea mays L.) seedlings. Plant Physiol **99**: 886–894

Azaizeh H, Steudle E (1991) Effects of salinity on water transport of excised maize (Zea mays L.) roots. Plant Physiol **97**: 1136–1145

Bolarín MC, Fernández FG, Cruz V, Cuartero J (1991) Salinity tolerance in four wild tomato species using vegetative yield-salinity response curves. J Am Soc Hortic Sci **116**: 286–290

Bourbia I, Pritzkow C, Brodribb TJ (2021) Herb and conifer roots show similar high sensitivity to water deficit. Plant Physiol **186**: 1908–1918

Boursiac Y, Chen S, Luu D-T, Sorieul M, van den Dries N, Maurel C (2005) Early effects of salinity on water transport in Arabidopsis roots. Molecular and cellular features of aquaporin expression. Plant Physiol **139**: 790–805

Brodribb TJ, Powers J, Cochard H, Choat B (2020) Hanging by a thread? Forests and drought. Science **368**: 261–266

Cai G, Ahmed MA, Abdalla M, Carminati A (2022) Root hydraulic phenotypes impacting water uptake in drying soils. Plant Cell Environ **45**: 650–663

Cai G, Ahmed MA, Dippold MA, Zarebanadkouki M, Carminati A (2020a) Linear relation between leaf xylem water potential and transpiration in pearl millet during soil drying. Plant Soil **447**: 565–578

Cai G, Ahmed MA, Reth S, Reiche M, Kolb A, Carminati A (2020b) Measurement of leaf xylem water potential and transpiration during soil drying using a root pressure chamber system. Acta Hortic 131–138

Cai G, Carminati A, Abdalla M, Ahmed MA (2021) Soil textures rather than root hairs dominate water uptake and soil–plant hydraulics under drought. Plant Physiol **187**: 858–872

Carminati A, Ahmed MA, Zarebanadkouki M, Cai G, Lovric G, Javaux M (2020) Stomatal closure prevents the drop in soil water potential around roots. New Phytologist **226**: 1541–1543

Carminati A, Javaux M (2020) Soil rather than xylem vulnerability controls stomatal response to drought. Trend Plant Sci **25**: 868–880

Carminati A, Passioura JB, Zarebanadkouki M, Ahmed MA, Ryan PR, Watt M, Delhaize E (2017) Root hairs enable high transpiration rates in drying soils. New Phytologist **216**: 771–781

Chaves MM, Flexas J, Pinheiro C (2009) Photosynthesis under drought and salt stress: regulation mechanisms from whole plant to cell. Ann Bot **103**: 551–560

Choat B, Brodribb TJ, Brodersen CR, Duursma RA, López R, Medlyn BE (2018) Triggers of tree mortality under drought. Nature **558**: 531–539

Deery DM, Passioura JB, Condon JR, Katupitiya A (2013) Uptake of water from a Kandosol subsoil. II. Control of water uptake by roots. Plant Soil **368**: 649–667

Donovan L, Linton M, Richards J (2001) Predawn plant water potential does not necessarily equilibrate with soil water potential under well-watered conditions. Oecologia **129**: 328–335

Draye X, Kim Y, Lobet G, Javaux M (2010) Model-assisted integration of physiological and environmental constraints affecting the dynamic and spatial patterns of root water uptake from soils. J Exp Bot **61**: 2145–2155

Fricke W, Akhiyarova G, Veselov D, Kudoyarova G (2004) Rapid and tissue-specific changes in ABA and in growth rate in response to salinity in barley leaves. J Exp Bot **55**: 1115–1123

Fricke W, Bijanzadeh E, Emam Y, Knipfer T (2014) Root hydraulics in salt-stressed wheat. Functional Plant Biol **41**: 366–378

Gupta RK, Verma SK (1985) Hydraulic conductivity of a swelling clay in relation to irrigation water quality. CATENA **12**: 121–127

Haber-Pohlmeier S, Vanderborght J, Pohlmeier A (2017) Quantitative mapping of solute accumulation in a soil-root system by magnetic resonance imaging. Water Resource Res **53**: 7469–7480

Hamza M, Anderson S, Aylmore G (2001) Studies of soil water drawdown by single radish roots at decreasing soil water content using computer-assisted tomography. Soil Res **39**: 1387–1396

Hamza M, Aylmore G (1992a) Soil solute concentration and water uptake by single lupin and radish plant roots - I. Water extraction and solute accumulation. Plant Soil **145**: 187–196

Hamza MA, Aylmore LAG (1992b) Soil solute concentration and water uptake by single lupin and radish plant roots. Plant Soil **145**: 197–205

Harper RJ, Dell B, Ruprecht JK, Sochacki SJ, Smettem KRJ (2021) Chapter 7 - Salinity and the reclamation of salinized lands. *In* JA Stanturf, MA Callaham, eds, Soils and Landscape Restoration. Academic Press, Cambridge, MA, pp 193–208

Hayat F, Ahmed MA, Zarebanadkouki M, Cai G, Carminati A (2019) Measurements and simulation of leaf xylem water potential and root water uptake in heterogeneous soil water contents. Adv Water Resource **124**: 96–105

Hayat F, Ahmed MA, Zarebanadkouki M, Javaux M, Cai G, Carminati A (2020) Transpiration reduction in maize (Zea mays L) in response to soil drying. Front Plant Sci **10**: 1695

Helliwell JR, Sturrock CJ, Mairhofer S, Craigon J, Ashton RW, Miller AJ, Whalley WR, Mooney SJ (2017) The emergent rhizosphere: imaging the development of the porous architecture at the root-soil interface. Sci Rep **7**: 1–10

Hopmans JW, Qureshi AS, Kisekka I, Munns R, Grattan SR, Rengasamy P, Ben-Gal A, Assouline S, Javaux M, Minhas PS, et al. (2021) Chapter one - critical knowledge gaps and research priorities in global soil salinity. *In* DL Sparks, ed, Advances in Agronomy. Academic Press, Cambridge, MA, pp 1–191

Javaux M, Carminati A (2021) Soil hydraulics affect the degree of isohydricity. Plant Physiol **186**: 1378–1381

Jorda H, Perelman A, Lazarovitch N, Vanderborght J (2018) Exploring osmotic stress and differences between soil–root interface and bulk salinities. Vadose Zone J **17**: 170029

Klopp HW, Daigh ALM (2020) Soil hydraulic model parameters and methodology as affected by sodium salt solutions. Soil Sci Soc Am J **84**: 354–370

Kumar P, Sharma PK (2020) Soil salinity and food security in India. Front Sustain Food Syst **4**: 174

Landsberg JJ, Fowkes ND (1978) Water movement through plant roots. Ann Bot **42**: 493–508

Li W, Zhang H, Wang W, Zhang P, Ward ND, Norwood M, Myers-Pigg A, Zhao C, Leff R, Yabusaki S, et al. (2021) Changes in carbon and nitrogen metabolism during seawater-induced mortality of Picea sitchensis trees. Tree Physiol **41**: 2326–2340

van Lier QJ, van Dam JC, Metselaar K (2009) Root water extraction under combined water and osmotic stress. Soil Sci Soc Am J **73**: 862–875

van Lier QJ, Metselaar K, van Dam JC (2006) Root water extraction and limiting soil hydraulic conditions estimated by numerical simulation. Vadose Zone J **5**: 1264–1277

Liu Y, Kumar M, Katul GG, Feng X, Konings AG (2020) Plant hydraulics accentuates the effect of atmospheric moisture stress on transpiration. Nat Clim Chang **10**: 691–695

Moldrup P, Olesen T, Komatsu T, Schjønning P, Rolston D e. (2001) Tortuosity, diffusivity, and permeability in the soil liquid and gaseous phases. Soil Sci Soc Am J **65**: 613–623

Munns R (1985) Na +, K + and Cl– in Xylem sap flowing to shoots of NaCl-treated barley. J Exp Bot **36**: 1032–1042

Munns R (2002) Comparative physiology of salt and water stress. Plant Cell Environ **25**: 239–250

Munns R, Day DA, Fricke W, Watt M, Arsova B, Barkla BJ, Bose J, Byrt CS, Chen Z-H, Foster KJ, et al. (2020a) Energy costs of salt tolerance in crop plants. New Phytologist **225**: 1072–1090

Munns R, Gilliham M (2015) Salinity tolerance of crops – what is the cost? New Phytologist **208**: 668–673

Munns R, Guo J, Passioura JB, Cramer GR (2000) Leaf water status controls day-time but not daily rates of leaf expansion in salt-treated barley. Funct Plant Biol **27**: 949–957

Munns R, Passioura J (1984) Effect of prolonged exposure to nacl on the osmotic pressure of leaf xylem sap from intact, transpiring barley plants. Funct Plant Biol **11**: 497

Munns R, Passioura JB, Colmer TD, Byrt CS (2020b) Osmotic adjustment and energy limitations to plant growth in saline soil. New Phytologist **225**: 1091–1096

Munns R, Tester M (2008) Mechanisms of salinity tolerance. Ann Rev Plant Biol **59**: 651–681

Passioura JB (1980) The transport of water from soil to shoot in wheat seedlings. J Exp Bot **31**: 333–345

Passioura JB (2020) Translational research in agriculture. Can we do it better? Crop Pasture Sci **71**: 517–528

Perelman A, Jorda H, Vanderborght J, Lazarovitch N (2020a) Tracing root-felt sodium concentrations under different transpiration rates and salinity levels. Plant Soil **447**: 55–71

Perelman A, Lazarovitch N, Vanderborght J, Pohlmeier A (2020b) Quantitative imaging of sodium concentrations in soil-root systems using magnetic resonance imaging (MRI). Plant Soil **454**: 171–185

Peters A, Iden SC, Durner W (2015) Revisiting the simplified evaporation method: identification of hydraulic functions considering vapor, film and corner flow. J Hydrol **527**: 531–542

Rodriguez-Dominguez CM, Brodribb TJ (2020) Declining root water transport drives stomatal closure in olive under moderate water stress. New Phytologist **225**: 126–134

Rouphael Y, Kyriacou MC, Colla G (2018) Vegetable grafting: a toolbox for securing yield stability under multiple stress conditions. Front Plant Sci **8**: 2255 doi: 10.3389/fpls.2017.02255

Schröder N, Lazarovitch N, Vanderborght J, Vereecken H, Javaux M (2014) Linking transpiration reduction to rhizosphere salinity using a 3D coupled soil-plant model. Plant Soil **377**: 277–293

Sperry JS, Love DM (2015) What plant hydraulics can tell us about responses to climate-change droughts. New Phytologist **207**: 14–27

Stirzaker RJ, Passioura JB (1996) The water relations of the root–-soil interface. Plant Cell Environ **19**: 201–208

Tuller M, Or D, Dudley LM (1999) Adsorption and capillary condensation in porous media: liquid retention and interfacial configurations in angular pores. Water Resource Res **35**: 1949–1964

Tyerman SD, Oats P, Gibbs J, Dracup M, Greenway H (1989) Turgor-volume regulation and cellular water relations of Nicotiana tabacum roots grown in high salinities. Funct Plant Biol **16**: 517–531

Vadez V (2014) Root hydraulics: the forgotten side of roots in drought adaptation. Field Crops Res **165**: 15–24

Wan X (2010) Osmotic effects of NaCl on cell hydraulic conductivity of corn roots. Acta Biochim Biophys Sin **42**: 351–357

van Zelm E, Zhang Y, Testerink C (2020) Salt tolerance mechanisms of plants. Ann Rev Plant Biol **71**: 403–433

Zhang H, Li X, Wang W, Pivovaroff AL, Li W, Zhang P, Ward ND, Myers-Pigg A, Adams HD, Leff R, et al. (2021) Seawater exposure causes hydraulic damage in dying Sitka-spruce trees. Plant Physiol **187**: 873–885